PROJECT-BASED LEARNING

PHASE-II EVALUATION

GRID RESCUE

Team ID: T091

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026–27

Team & Mentor Details

Team Information

Team ID: T091
Semester: 3rd
Section: XX
Domain: *B.Tech 2nd-Year DSA+OOPS*

Team Members

1. *Tejasv Singh Nagarkoti – 2510010772*
2. *Tanishq Vashisth – 2510010163*
3. *Tanishka Sahu – 2510017468*
4. *Riddhiman Sandilya – 2510370168*

Mentor

Dr.Siddhant Thapliyal

Interactions completed: *4*

Brief Discussion: Introduce the team, project domain, mentor, and responsibilities of the team members.

Phase-I Recap & Feedback Incorporated

Phase-I Status

Proposed a power-grid failure simulation and Discussed project idea and working.

Feedback Received

- *Discuss the major technical challenges*
- *Explain the approach for GUI development2*
- *Clarify the role of DSA algorithms*

Changes Incorporated

- *Identified challenges in project implementation*
- *Explored GUI development using ImGui*
- *Refined the approach to graph algorithms*

Brief Discussion: Explain the major feedback received during Phase-I and clearly demonstrate how it has been incorporated.

Problem Statement & Finalized Objectives

Problem Statement

Developing a power-grid simulation to study transmission-line failures and their effect on network connectivity.

Finalized Objectives

1. *Represent grid components using C++ classes.*
2. *Model connections between grid components.*
3. *Simulate transmission-line failures.*
4. *Explore affected nodes and alternate routes using graph algorithms.*

Brief Discussion: Avoid repeating the complete Phase-I presentation. Focus on the finalized problem, scope and measurable objectives.

Updated System Architecture / Workflow



Major Components

- *Grid Component Classes*
- *Graph Algorithm Modules*
- *FailureManager PowerManager*

Changes Since Phase-I

- *C++ backend implementation completed*
- *Failure and recovery simulation working*
- *Working terminal-based simulation*

Brief Discussion: Present the updated architecture and explain how individual modules communicate with each other.

Module Design & Functional Requirements

Major Modules

1. *Grid Management Module*
2. *Graph Algorithm Module*
3. *Failure Management Module*
4. *Power Management Module*

Functional Requirements

- *Create and display grid components.*
- *Apply and undo component failures.*
- *Identify affected areas and alternate routes.*
- *Redistribute power by consumer priority.*

Module Dependency

GridManager manages grid components, Graph and Algorithms check connectivity, FailureManager handles failures, and PowerManager redistributes power. Main.cpp coordinates the entire program.

Technology Stack & Development Methodology

Technology Stack

- Programming: *C / C++ / Other*
- Database:
- Framework: *Dear ImGui*
- Development Tool: *VS code*
- Version Control: *Git / GitHub*

Development Methodology

1. Requirement Analysis
2. System Design
3. Module Development
4. Integration
5. Testing & Validation

Brief Discussion: Explain why the selected technologies are appropriate and how the team is following the development methodology.

Implementation of PBL Course Concepts

Course	Concepts Actually Implemented	Project Module
Data Structures in C	*Arrays, Linked Lists, Trees, Graphs, Hashing, etc.*	*Graph, Algorithm, PowerManager*
OOPs with C++	*Classes, Inheritance, Polymorphism, STL, etc.*	*GridManager, PowerStation, Substation, Consumer, TransmissionLine*
Operating Systems		–
DBMS		–

Important: Mention only those concepts that have been **actually implemented** in the project.

Brief Discussion: Students should explain where and how each course concept is technically implemented.

Development & Implementation Progress

Module	Status	Work Completed
Grid Management	Completed	*Grid components and connections*
Failure,GraphAlgorithms	Completed	*Priority-based power redistribution*
Power Management	Completed	*Failure analysis and alternate routest*
GUI Interface	Pending	*ImGui visualisation*

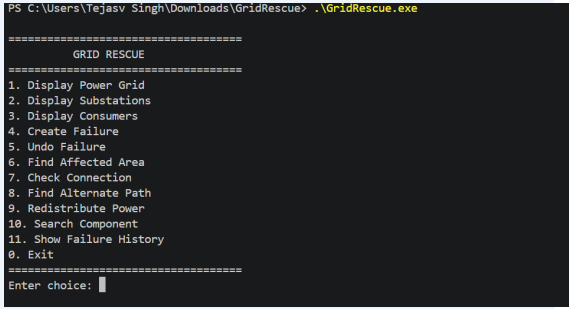
Overall Project Completion: *70%*

Brief Discussion: Clearly differentiate between completed, in-progress and pending development. Do not present proposed work as completed work.

Working Prototype / Demonstration

Implemented Features

- *Shows power stations, substations, consumers, and their connections.*
- *Simulates grid failures and identifies affected areas.*
- *Uses BFS, DFS, and Dijkstra's algorithm to check connections and find alternate paths.*
- *Prioritizes important consumers.*



Testing, Results & Validation

Test / Module	Expected Result	Observed Result	Status
Program Execution	*Main menu should appear.*	*Main menu displayed successfully.*	Pass
Test 2: Grid Display	*Grid components displayed.*	*Grid display works through the console.*	Pass
Test 3: GUI Simulation	*Grid works through GUI.*	*not working yet*	*Pending*

Results / Validation

The program launches successfully and displays the grid through the console. Basic functionality has been tested, while GUI simulation is still under development. Response time and algorithm accuracy have not yet been measured.

Brief Discussion: Results should be supported by actual testing or experimental evidence rather than assumptions.

GitHub Repository & Development Activity

Repository Information

- GitHub Link: *https://github.com/tejasvsinghnagarkoti-234/Gridrescue.git*
- README.md: *Updated*
- Contributors: *All Members Added*
- Repository Status: *Updated*
- Branches: *1(main)*

Development Activity

- *Regular commits maintained*
- *Meaningful commit messages*
- *Individual contribution visible*
- *Latest implementation uploaded*
- *Documentation maintained*

Requirement: All team members must be contributors and must **commit individually using their own GitHub accounts**.

Brief Discussion: Show repository activity, README, contributor list and commit history during the

Individual Contribution

Member	Role	Technical Contribution	GitHub Commits
Tejasv Singh Nagarkoti	*Role*	*Graph Data Structure and Traversal Algorithms (BFS + DFS)*	*18*
Tanishq Vashisth	*Role*	*Dijkstra's Algorithm and Power Redistribution Engine*	*18*
Tanishka Sahu	*Role*	*OOP Data Architecture, Entity Modeling and Display System*	*23*
Riddhiman Sandilya	*Role*	*Failure Simulation, Undo System and Grid Orchestration*	*24*

Individual Evaluation

Every member should be prepared to explain and demonstrate their own technical contribution to the project.

Challenges, Remaining Work & Phase-III Roadmap

Challenges Faced

- *Technical:Implementing graph algorithms for grid failures and pathfinding.*
- *Integration:Connecting the console logic with the GUI.*
- *Implementation:Handling failed components.*

Solutions / Learning

- *Improved understanding of graph algorithms.*

Phase-III Roadmap

1. *Complete the GUI simulation.*
2. *Integrate all modules.*
3. *Test and improve functionality.*
4. *GUI simulation*
5. *Complete Documentation*

Brief Discussion: Explaining the technical challenges and provide a realistic roadmap for completing the

Expected Outcomes & References

Expected Final Outcomes

- *Working power-grid simulation*
- *Testing algorithm performance*
- *Faster detection of affected consumers*
- *Alternative routes during line failures*
- *Scope for larger grid simulations*

References

1. *IEEE Xplore Digital Library — Resources on Power System Reliability and Fault Analysis.*
2. *Programiz — Graph Data Structure*
3. *E.Balagurusamy*
4. *Jcode book*
5. *GridRescue Project Repository — GitHub*

Thank You
Questions & Suggestions